Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/01/2006
PI Name:	Lysakowski, Anna Ph.D.		
Project Title:	An Ultrastructural Study of Development in Vestibular Hair Cells, and their Cellular Responses to Hypergravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	60612	**Congressional District:**	7
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
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	The broad long-term objective of this research is to understand sensory processing in the vestibular system. Specific aims are: (1) to provide a detailed ultrastructural analysis of regional variations in cellular and synaptic architecture in the utricular macula of the developing and adult mammal, and (2) to study the influence of hypergravity on synaptic activity, synaptic plasticity and aging in the rat utricular macular hair cell. In previous work, we have described regional variations in synaptic innervation of the crista. Similar studies are underway for the utricular macula in the developing, adult, and aged mammal. These studies will provide baseline data for future flight studies. In addition to this, we are examining the effect of hypergravity conditions upon certain features of synaptic innervation, namely plasticity of synaptic innervation and the influence of hypergravity upon calyceal invaginations (one possible indicator of synaptic activity), and the cellular structure of hair cells, particularly in aged animals. Some of the cellular and synaptic features that vary regionally are likely to be a physiological response to stimulation We are defining the anatomical limits of the		

Task Description:	striolar region of the utricular and saccular maculae and presented results on the saccular macula this past year. In these studies, we have both qualitatively and quantitatively delineated the borders of the striola using a molecular marker, the calcium-binding protein calretinin. This marker stains pure calyx afferents, which are restricted to the striola. Using this marker, we have done a comparative survey of several of the rodents commonly used in inner ear studies: mouse, rat, gerbil, guinea pig, chinchilla and squirrel. Knowing the precise borders and the proportions of calyx afferents contained in the striola will help us in our hypergravity studies of synaptic activity in hair cells in both normal adult and aged animals. This study has been submitted for publication. In addition to the studies on the striolar morphometry, we also presented preliminary results on our hypergravity experiments. Multiple samples were taken from each rat utricular macula. Each sample spanned the entire sensory epithelium and included material from all three regions: striola, juxtastriola, and medial and lateral extrastriola. Dissector counts of synaptic ribbons and calyceal invaginations (another possible indicator of synaptic activity) were made as described previously in a study of the chinchilla crista ampullaris (Lysakowski & Goldberg, 1997). Preliminary results from utricular macula indicate that there are slightly lower numbers of synaptic ribbons per hair cell in the rat and that these numbers do not vary by region, but that calyceal invaginations are more numerous in the striola, compared to the extrastriola. Synaptic ribbons in type II hair cells are larger and occur in clusters more often in the striola compared to the extrastriola, similar to our results in the crista. Type II hair cells also have more ribbons per hair cell than do type I hair cells (about 15 vs 8). Preliminary results of experiments in four samples from spun adult animals indicate that there were more synaptic ribbons and calyceal invaginations in the excited striola than in the inhibited striola, in type II hair cells. A similar effect is now seen in the extrastriola for type I and type II hair cells. We are continuing to analyze more samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With the background information on the normal synaptic innervation patterns that we now have, we can determine how this pattern is altered by linear acceleration, the appropriate stimulus for the utriculus. This project, which focuses on how cells respond to both acute and long-term variations in gravity, is directly relevant to one of the stated goals of NASA's Space Biology Program for Cellular and Molecular Biology. This goal is "to elucidate the mechanisms by which cells... respond to gravitational perturbations" (from NASA's Research Announcement: Research Opportunities in Space Life Sciences, June, 1998). Up till now, the scientific literature offers no answer to the question of whether or not there is a critical period for vestibular sensory development. Our preliminary data suggest that there is. This project will provide baseline data for future studies on the influence of hypergravity and microgravity on animal growth and development. It is directly relevant to one of the stated goals of NASA's Space Biology Program for Developmental Biology. This goal is "to determine the mechanisms by which gravity induces normal development and function" (from NASA's Research Announcement: Research Opportunities in Space Life Sciences, June, 1998). A second developmental, and cellular, change that we are investigating as an unexpected result of our studies is that of altered cytoskeletal changes in the hair cells of aged animals due to hypergravity conditions. This project is directly relevant to another of the stated goals of NASA's Space Biology Program for Cellular and Molecular Biology, which is to "understand how direct or indirect effects of altered gravitational forces impact...cytoskeletal structure and function" (from NASA's Research Announcement: Research Opportunities in Space Life Sciences, June, 1998).
Task Progress:	No progress report received this reporting period.
Bibliography Type:	Description: (Last Updated: 07/29/2019)